

MODULE 1

INTRODUCTION TO SOLID WASTE MANAGEMENT

Definition, classification, need and global perspective of solid waste management. Policies and legislative frameworks, government initiatives on solid waste management. Integrated solid waste management and concepts of 3R's, Role of stakeholders

SOLID WASTE

DEFINITION

Waste arising from human and animals activities that are normally solids and discarded as useless and unwanted or

Heterogeneous mass thrown away from residencies, commercial activities as well as more homogeneous accumulations of a single industry activity

NEED FOR SWM

- **Protects the Environment:** Helps prevent pollution of air, water, and land.
- **Reduces Health Risks:** Keeps neighbourhoods clean and reduces diseases caused by poor waste disposal.
- **Saves Resources:** Recycles materials like plastic, paper, and metals, reducing the need for new raw materials.
- **Cuts down Energy Use:** Using recycled materials saves energy compared to making new products from scratch.
- **Reduces Landfill Waste:** Limits the amount of waste going to overflowing landfills.
- **Helps Fight Climate Change:** Properly managing waste reduces harmful greenhouse gas emissions.
- **Creates Jobs:** Recycling and waste management industries provide job opportunities.
- **Supports Sustainable Growth:** Encourages the efficient use of resources and a cleaner, greener future.
- **Keeps Places Beautiful:** Prevents littering and makes cities and towns more attractive and liveable.
- **Meets Legal Standards:** Helps cities and companies follow government rules on waste disposal, avoiding fines.

CAUSES OF SW

- Advanced in technology
- Use and throw culture
- Advantages of packaging technology
- Increase in urbanization
- Over population
- Affluence- High purchasing capacity

GLOBAL PERSPECTIVE OF SWM

Solid waste management is a pressing environmental and public health issue across the globe. Rapid urbanization, industrialization, and population growth have led to a significant increase in waste generation, particularly in urban areas. Here's an overview of SWM practices, challenges, and advancements from a global perspective:

1. Waste Generation Trends

- **Developed Countries:** Industrialized nations like the USA, European countries, and Japan produce vast quantities of waste, primarily municipal solid waste (MSW). Due to high consumption rates, the waste per capita is significantly higher compared to developing countries. However, these nations also have more efficient waste management systems.
- **Developing Countries:** Countries in Asia, Africa, and Latin America are experiencing a surge in waste generation due to urbanization and industrial growth. Waste collection is often inefficient, and a significant portion of waste is not properly treated.

2. Key Challenges

- **Waste Segregation:** Many countries struggle with proper waste segregation at the source (household or industry), which is crucial for recycling and composting.
- **Recycling Rates:** Global recycling rates vary widely. Developed countries like Germany and South Korea have high recycling rates, whereas many developing countries lack proper infrastructure and public awareness.
- **Landfills:** The predominant method of disposal in many parts of the world is landfilling, which often leads to environmental issues such as land contamination and methane emissions. In low-income countries, open dumping is more common.
- **Plastic Waste:** Plastic pollution is a global crisis, especially in marine environments. Countries with inadequate waste management infrastructure contribute disproportionately to this issue.
- **E-waste:** The management of electronic waste (e-waste) is a growing concern globally, with inadequate recycling practices leading to environmental hazards.

3. Best Practices

- **Circular Economy Approach:** Countries like Japan and some in Europe are adopting a circular economy model, where waste materials are reintroduced into the production cycle, reducing waste.
- **Extended Producer Responsibility (EPR):** Several nations are implementing EPR policies, making manufacturers responsible for the disposal or recycling of their products, especially in electronics and packaging industries.
- **Zero Waste Initiatives:** Some cities (e.g., San Francisco, Kamikatsu in Japan) are aiming to eliminate all waste sent to landfills by maximizing recycling, composting, and reducing waste generation.

4. Technological Advances

- **Waste-to-Energy (WtE):** Countries like Sweden and Denmark are leaders in converting waste to energy through incineration, reducing landfill use and generating electricity. *(Sweden has numerous modern incineration plants that convert household waste into electricity and heat through combustion. Sweden imports waste from other European*

countries, such as the UK and Norway. This imported waste is incinerated to produce energy, making Sweden a key player in European waste management.)

- **Anaerobic Digestion:** Widely used in Europe, this technology converts organic waste into biogas, which can be used for electricity or heat, contributing to renewable energy generation.
- **Advanced Recycling Technologies:** Innovations in recycling technology (e.g., chemical recycling for plastics) are gaining traction, particularly in developed nations.

5. Global Policies and Agreements

- **Sustainable Development Goals (SDGs):** The UN's SDG 12 (Responsible Consumption and Production) targets reducing waste generation through prevention, reduction, recycling, and reuse.
- **Basel Convention:** This international treaty aims to control the transboundary movement of hazardous wastes, including e-waste, particularly from developed to developing countries.
- **Paris Agreement:** Indirectly, waste management contributes to the goals of the Paris Agreement, as better waste practices reduce greenhouse gas emissions from landfills and waste-to-energy facilities. (*The Paris Agreement is an international treaty that aims to reduce the effects of climate change by limiting greenhouse gas emissions and helping countries adapt to the consequences of climate change. The agreement was adopted in December 2015 at the United Nations Climate Change Conference (COP21) in Paris, France and came into effect on November 4, 2016.*)

6. Case Studies

- **Germany:** Known for its advanced waste management system, Germany leads the world in recycling and waste recovery efforts, with over 65% of its waste being recycled.
- **Sweden:** Sweden's WtE plants incinerate almost 50% of the country's waste, with only 1% of household waste ending up in landfills.
- **India:** India faces severe challenges in SWM due to rapid urbanization. However, initiatives like the Swachh Bharat Mission aim to improve sanitation and waste management practices.

Integrated Solid Waste Management (ISWM)

1. Definition:

Integrated Solid Waste Management (ISWM) refers to the comprehensive management of solid waste that encompasses the collection, transportation, treatment, and disposal of waste materials. It aims at minimizing the environmental impact while maximizing resource recovery.

2. Objectives:

- Minimize waste generation.
- Maximize resource recovery and recycling.
- Ensure safe disposal of residual waste.
- Protect human health and the environment.

3. Hierarchy of Waste Management:

ISWM follows a waste management hierarchy, often referred to as the "3 Rs" which prioritize waste management options:

- **Reduce:** Minimizing waste generation at the source.
- **Reuse:** Using materials again in their current form.
- **Recycle:** Converting waste materials into new products.

Other strategies include:

- **Recovery:** Extracting energy from waste (e.g., waste-to-energy plants).
- **Disposal:** Safe disposal of waste in landfills or incineration, which should be the last resort.

4. **Components of ISWM:**

ISWM involves several components working together:

- **Waste Generation:** Source control and reduction.
- **Waste Segregation:** Separating waste into categories such as organic, recyclable, hazardous, etc.
- **Waste Collection:** Efficient systems for collecting waste from residential, commercial, and industrial areas.
- **Waste Transportation:** Safe and cost-effective transport of waste to treatment facilities.
- **Waste Treatment:** Different treatment methods like composting (for organic waste), incineration, recycling, and anaerobic digestion.
- **Waste Disposal:** Disposal of non-recyclable and non-recoverable waste in landfills or through incineration.

5. **Key Methods in ISWM:**

- **Landfilling:** Disposal of waste in landfills, which requires proper management to prevent groundwater contamination.
- **Composting:** Organic waste is biologically decomposed under controlled conditions to produce compost.
- **Recycling:** Recovery of materials such as plastics, metals, glass, and paper, and reprocessing them into new products.
- **Waste-to-Energy (WtE):** Energy is extracted from the combustion of waste materials.
- **Incineration:** Burning waste at high temperatures to reduce volume and destroy hazardous substances.

6. **Environmental and Economic Benefits:**

- Reduces the volume of waste sent to landfills and incineration.
- Reduces greenhouse gas emissions.
- Generates energy from waste.
- Recovers valuable resources, thus reducing the demand for virgin materials.
- Creates job opportunities in the waste management and recycling sectors.

Concepts of the 3 R's: Reduce, Reuse, and Recycle

The 3 R's—**Reduce, Reuse, and Recycle**—are fundamental principles for sustainable waste management. They aim to minimize waste generation and promote efficient use of resources.

1. Reduce

Definition: Minimizing the amount of waste generated by reducing consumption at the source.

Key Concepts:

- **Preventing waste** before it is created.
- Encouraging efficient resource use in production and daily activities.
- **Examples:**

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- Buying products with minimal packaging.
 - Using energy-efficient appliances to reduce energy consumption.
 - Opting for digital alternatives instead of printed materials.
- **Environmental Impact:** Reducing the quantity of waste leads to a decrease in pollution, conservation of natural resources, and reduced energy use.

Actions to Reduce:

- Choose durable, long-lasting products.
- Avoid single-use items (e.g., plastic bags, disposable cups).
- Support companies that practice sustainable manufacturing.

2. Reuse

Definition: Using products or materials more than once in their original form before disposing of them.

Key Concepts:

- Extending the lifecycle of items by repurposing or refurbishing them.
 - Avoiding the need for new products,
- which reduces resource extraction and manufacturing impacts.
- **Examples:**
 - Reusing glass jars for storage.
 - Repairing broken appliances or electronics instead of replacing them.
 - Donating old clothes, furniture, or household items to others.
 - **Environmental Impact:** Reuse conserves raw materials and reduces energy use associated with manufacturing new products, while also cutting down on waste.

Actions to Reuse:

- Opt for reusable containers, shopping bags, and bottles.
- Refurbish old furniture or electronics.
- Share or donate items that are no longer needed.

3. Recycle

Definition: Converting waste materials into new products by reprocessing them.

Key Concepts:

- Recycling transforms waste into raw materials that can be used to create new products, reducing the need for virgin resources.
- Recycling helps conserve resources such as metal, paper, and plastic and minimizes the amount of waste sent to landfills.
- **Examples:**
 - Recycling paper to make new paper products.
 - Melting down aluminum cans to create new cans.
 - Transforming plastic bottles into synthetic fibers for clothing.
- **Environmental Impact:** Recycling reduces landfill waste, saves energy, and conserves natural resources like timber, water, and minerals.

Actions to Recycle:

- Separate recyclable materials (paper, plastics, metals) from general waste.
- Support products made from recycled materials.
- Use local recycling services for appropriate disposal.

Benefits of the 3 R's:

- **Conservation of Resources:** By reducing, reusing, and recycling, fewer raw materials are extracted from the environment.
- **Energy Savings:** Recycling and reusing products require less energy compared to producing new items from raw materials.
- **Pollution Reduction:** Decreasing waste generation and diverting waste from landfills reduces pollution and greenhouse gas emissions.
- **Economic Savings:** Reducing consumption and reusing items can lead to cost savings, both for individuals and industries.

Role of Stakeholders in Solid Waste Management (SWM)

Effective **Solid Waste Management (SWM)** requires the involvement of multiple stakeholders, each playing a specific role in the planning, implementation, and operation of waste management systems. The success of an integrated SWM system depends on the collaboration and coordination among various groups.

1. Government (Local, State, and National Authorities)

- **Role:** Policy formulation, regulation, oversight, and provision of essential infrastructure.
- **Key Responsibilities:**
 - **Regulation:** Develop and enforce laws, regulations, and policies related to waste management (e.g., waste segregation mandates, landfill operation standards, recycling incentives).
 - **Infrastructure:** Provide and maintain essential infrastructure like collection services, transfer stations, landfills, recycling plants, and waste-to-energy facilities.
 - **Public Awareness:** Run campaigns to educate the public on waste reduction, segregation, and recycling.
 - **Funding & Budgeting:** Allocate budgets and resources for solid waste management projects.
 - **Monitoring & Evaluation:** Monitor and assess the performance of waste management systems and ensure compliance with environmental standards.

2. Municipalities and Local Bodies

- **Role:** Directly responsible for the collection, transport, and disposal of waste at the community level.
- **Key Responsibilities:**
 - **Waste Collection and Transport:** Organize efficient waste collection services (door-to-door collection, public bins) and transportation to treatment or disposal facilities.
 - **Enforcement:** Enforce local waste management regulations, including proper waste segregation and prevention of illegal dumping.
 - **Waste Treatment Facilities:** Establish and manage local waste treatment plants (e.g., composting centres, recycling plants).

- **Community Engagement:** Work closely with communities to raise awareness about waste segregation and encourage sustainable practices.

3. Private Sector (Waste Management Companies, Industries, and Businesses)

- **Role:** Contribute through innovation, service delivery, and investment in waste management infrastructure.
- **Key Responsibilities:**
 - **Collection and Disposal:** Private companies may be contracted to handle specific tasks such as waste collection, transportation, and recycling.
 - **Recycling and Resource Recovery:** Businesses can invest in recycling technologies and waste-to-energy solutions to extract value from waste.
 - **Corporate Social Responsibility (CSR):** Engage in environmentally friendly practices such as reducing industrial waste, promoting recycling, and funding community waste projects.
 - **Innovation and Technology:** Develop new technologies and approaches to improve waste management efficiency, such as smart bins, tracking systems, and automated sorting.

4. Non-Governmental Organizations (NGOs) and Community-Based Organizations (CBOs)

- **Role:** Raise awareness, advocate for environmental justice, and assist in mobilizing local communities for sustainable waste management.
- **Key Responsibilities:**
 - **Public Education and Advocacy:** Promote campaigns on waste reduction, proper segregation, recycling, and composting practices at the grassroots level.
 - **Policy Advocacy:** Influence policies related to waste management and push for more sustainable and equitable practices.
 - **Capacity Building:** Train local authorities and communities on best practices for waste handling, recycling, and composting.
 - **Support for Informal Sector:** Advocate for the inclusion and welfare of informal waste collectors (e.g., rag-pickers) and help integrate them into formal waste management systems.

5. Informal Waste Sector (Waste Pickers, Scrap Dealers)

- **Role:** Play a crucial role in the collection, sorting, and recycling of waste materials, particularly in developing countries.
- **Key Responsibilities:**
 - **Material Recovery:** Informal waste workers recover valuable materials like paper, plastics, metals, and glass from waste streams, diverting them from landfills and facilitating recycling.
 - **Economic Contribution:** Create livelihoods for many low-income individuals through waste recovery and recycling activities.
 - **Partnerships with Municipalities:** In some cases, municipalities and private companies partner with informal waste collectors to formalize their activities and provide access to social security benefits and safer working conditions.

6. Households and Individuals

- **Role:** The primary generators of waste, individuals are critical to the success of waste management systems through participation in waste reduction, segregation, and recycling.
- **Key Responsibilities:**
 - **Waste Segregation:** Properly segregate waste into categories (organic, recyclable, non-recyclable) at the source.

- **Reduction of Waste:** Practice the 3 R's—reduce, reuse, and recycle—by minimizing waste generation, reusing items, and participating in recycling programs.
- **Community Participation:** Engage in local SWM initiatives, including clean-up drives, composting projects, and recycling campaigns.
- **Awareness and Education:** Stay informed about proper waste disposal practices and support environmentally responsible behavior in the community.

7. Academia and Research Institutions

- **Role:** Provide research and innovation to improve waste management practices and develop sustainable solutions.
- **Key Responsibilities:**
 - **Research and Development:** Conduct research on waste management technologies, environmental impact assessments, and new methods for resource recovery.
 - **Training and Capacity Building:** Offer education and training programs for students, professionals, and municipal staff on waste management practices.
 - **Policy Support:** Provide evidence-based recommendations to policymakers on improving SWM systems and policies.

8. Media

- **Role:** Raise awareness and educate the public on the importance of effective waste management.
- **Key Responsibilities:**
 - **Public Information Campaigns:** Broadcast information on waste management issues, best practices, and government initiatives.
 - **Advocacy and Reporting:** Highlight environmental and social issues related to poor waste management and advocate for change.

Indian Government Initiatives on Solid Waste Management (SWM)

India has been facing significant challenges related to solid waste management due to rapid urbanization, population growth, and industrial development. In response, the Indian government has introduced several policies and programs aimed at improving solid waste management and ensuring sustainability. Below are some of the key initiatives:

1. Swachh Bharat Mission (SBM) – Urban & Rural

- **Launched:** 2014
- **Objective:** The flagship initiative by the Indian government to ensure cleanliness and sanitation across India, including effective solid waste management.
- **Key Features:**
 - **Promotion of Waste Segregation:** Encouraging households to segregate waste into wet (organic) and dry (recyclable) categories.
 - **Door-to-Door Collection:** Promoting 100% door-to-door waste collection.
 - **Waste Processing:** Setting up composting units, biogas plants, and waste-to-energy (WtE) plants to treat solid waste and convert it into useful products such as compost, energy, or fuel.

- **Public Participation:** Emphasizing the importance of public participation through awareness campaigns and behavioural change initiatives.
- **Achievements:** Many cities have achieved improved waste collection efficiency, and several urban centers have initiated composting and recycling programs.
- **SBM 2.0 (2021):** Launched to further strengthen urban sanitation with a focus on sustainable waste management and 100% source segregation, scientific processing of all waste, and reducing landfilling.

2. Solid Waste Management Rules, 2016

- **Issued by:** Ministry of Environment, Forest and Climate Change (MoEFCC)
- **Objective:** A comprehensive set of rules aimed at regulating waste management activities across the country.
- **Key Provisions:**
 - **Source Segregation:** Mandates waste generators (households, institutions, and industries) to segregate waste into biodegradable, non-biodegradable, and hazardous categories.
 - **Extended Producer Responsibility (EPR):** Producers and brand owners are responsible for managing the waste generated from their products (e.g., plastics, electronics) at the end of their life cycle.
 - **Inclusion of the Informal Sector:** Encouraging local bodies to integrate waste pickers and the informal waste sector into formal waste management systems.
 - **Decentralized Waste Processing:** Promoting composting, biogas generation, and recycling at the local level.
 - **Penalties for Non-compliance:** Imposing penalties for not complying with the regulations regarding waste segregation, disposal, and treatment.

3. Plastic Waste Management Rules, 2016 (Amendments in 2021)

- **Objective:** To regulate the manufacture, sale, and disposal of plastic waste in the country.
- **Key Features:**
 - **Ban on Single-Use Plastics:** The rules ban the use of single-use plastic products such as plastic bags, straws, and cutlery (to be phased out by 2022).
 - **Extended Producer Responsibility (EPR):** Mandates producers, importers, and brand owners to ensure the collection and processing of plastic waste.
 - **Recycling and Reuse:** Encourages recycling and proper disposal of plastic waste to minimize its environmental impact.
 - **Plastic Waste Processing:** Promoting the development of recycling plants and facilities for processing plastic waste into useful products.

4. Waste-to-Energy (WtE) Initiatives

- **Objective:** Convert waste materials into energy, including electricity, heat, or fuel, through incineration or other waste-to-energy technologies.
- **Key Projects:**
 - **Setting up WtE Plants:** Several waste-to-energy plants have been established in urban centers such as Delhi, Hyderabad, and Bengaluru.
 - **Biogas and Composting:** The government promotes biogas production from organic waste and composting initiatives through decentralized waste management facilities.
 - **National Policy on Biofuels:** Encourages the conversion of urban waste into energy, especially in the form of biofuels like biogas.

5. National Green Tribunal (NGT) Orders

- **Role:** The NGT has played an active role in ensuring compliance with SWM regulations by issuing specific guidelines and orders related to waste management.
- **Key Orders:**
 - Mandating the closure of illegal dump sites and ensuring the scientific management of existing landfills.
 - Penalizing municipalities that fail to comply with waste management rules and regulations.
 - Directing state governments to develop comprehensive waste management plans and ensure proper waste segregation at the source.

6. Atal Mission for Rejuvenation and Urban Transformation (AMRUT)

- **Launched:** 2015
- **Objective:** Improving urban infrastructure and providing basic services such as sanitation and waste management in 500 cities across India.
- **Key Features:**
 - Focus on creating waste management systems in urban areas, including waste collection, segregation, transportation, and processing.
 - Encourages cities to establish waste processing plants, recycling units, and promote waste reduction at the source.
 - Provides financial assistance to municipalities for setting up solid waste management infrastructure.

7. Smart Cities Mission

- **Launched:** 2015
- **Objective:** To develop 100 smart cities across India with a focus on urban sustainability, including integrated solid waste management systems.
- **Key Features:**
 - Promotion of smart waste management systems using technology like smart bins, automated waste collection systems, and tracking waste transportation.

- Encouraging cities to adopt innovative waste treatment methods like decentralized composting and recycling facilities.
- Real-time monitoring of waste management through centralized command centers.

8. National Action Plan for Climate Change (NAPCC) – Mission on Sustainable Habitat

- **Objective:** Address urban waste management as part of a broader goal to promote sustainable urbanization in response to climate change.
- **Key Features:**
 - Encourages energy-efficient waste management processes, including waste-to-energy and biogas generation.
 - Promotes the reduction of greenhouse gas emissions by minimizing landfill use and promoting recycling and composting.
 - Supports the integration of SWM systems into urban planning as part of the National Mission on Sustainable Habitat.

9. Extended Producer Responsibility (EPR)

- **Objective:** Ensure that manufacturers and producers are responsible for the entire lifecycle of their products, especially concerning the management of post-consumer waste.
- **Key Features:**
 - EPR covers multiple waste streams like plastics, electronics (e-waste), and packaging waste.
 - Producers are required to set up collection and recycling systems for the waste generated from their products.
 - Encourages industries to design eco-friendly, recyclable products to minimize waste generation.

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SOLID WASTE MANAGEMENT

SUBJECT CODE : 10CV 757

PART-A

UNIT-I

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INTRODUCTION AND SOURCES

SOLID WASTE Solid wastes are all the wastes arising from human and animal activities that are normally solid and are discarded as useless or unwanted.

Solid wastes comprises of materials often reusable and may be considered as resource in another setting. It consists of

- Heterogeneous mass → From residences and commercial activities
- Homogeneous mass → From single industrial activity

SOLID WASTE MANAGEMENT solid waste management is a discipline associated with.

1. Control of generation
2. Storage
3. Collection
4. Transfer and transport
5. Processing
6. Utilisation and disposal of solid waste in most economical way in order to protect public health & natural environment

Local authorities are responsible for proper solid waste management practices. The main method of safe management

of waste includes

- * Segregation

- * Collection

- * Dumping

LAND POLLUTION

DEFINITION Land pollution means degradation or destruction of earth's surface and soil, directly or indirectly as a result of human activities.

OR

It refers to any physical or chemical alteration of land, rendering it incapable of beneficial use without treatment.

CAUSES OF LAND POLLUTION

Below are the sources of land pollution.

INCREASE IN URBANIZATION: Construction uses up forest land. More construction means increase in demand for raw materials like timber. This leads to exploitation and destruction of forests.

INCREASE IN AGRICULTURAL LAND: As human population grew there was a greater demand for food. This caused more land allocated to agriculture. Forests were cut down for this purpose which causes soil erosion.

Farmers often use highly toxic fertilizers and pesticides to get rid of insects, fungi and bacteria from their crops. By overuse of these chemicals, they result in contamination and poisoning of soil.

DOMESTIC WASTES: Every single day, tons and tons of domestic waste is dumped. If all these wastes are not disposed of properly, the damage they can do to environment and human kind can be dangerous.

INDUSTRIALIZATION: Industrial wastes are the effluents discharged from chemical industries, paper & pulp mills, tanneries, distilleries, refineries, pesticide and fertiliser, pharmaceutical industries, food processing industries, cement industries, mining, nuclear power plants etc....

MINING ACTIVITIES: Mining affects ground and surface waters, aquatic life, vegetation, soils, animals and the human health.

Mining gas and petroleum also pollutes the land. Digging is done to extract underground minerals which destroys top soil and contaminates area with toxic metals and chemicals.

OVERCROWDED LAND FILLS: Each house hold produces tonnes of ~~each~~ garbage each year. Garbage like aluminium, plastic, paper, wood, cloth etc... collected and sent to land filling which causes land pollution.

SEWAGE TREATMENT: Large amount of solid waste is left over after sewage has been treated, this waste is sent for land fill which pollutes land and environment.

EFFECTS OF LAND POLLUTION

SOIL POLLUTION: * Upper layer of the ~~soil~~ ^{soil} is damaged

* caused by overuse of chemical fertiliser

- Soil erosion

* Leads to loss of fertile land for agriculture, forest cover etc.

AIR POLLUTION: Increased wastes from landfills were burnt which leads to air pollution which can cause diseases.

CHANGE IN CLIMATIC PATTERNS:

* The deforestation leads to imbalance in rain cycle.

* Variations in climatic patterns leads to global warming, green house effect, floods etc...

ENVIRONMENTAL IMPACT:

- Global warming
- Green house effect
- Floods
- Uneven rainfall

- Increased temperature
- Acid rain
- Ecological imbalance.

EFFECT ON HUMAN HEALTH:

- Skin cancer
- Respiratory problems

} Due to toxic chemicals which can enter through foods & vegetables grown from polluted soil

EFFECT ON WILD LIFE:

- Loss of habitat and natural environment
- Migration of species

DISTRACTION TO TOURIST:

- * City loses its attraction as tourist destination as landfills
- * Leads to loss of revenue for government.

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SOLUTIONS FOR LAND POLLUTION

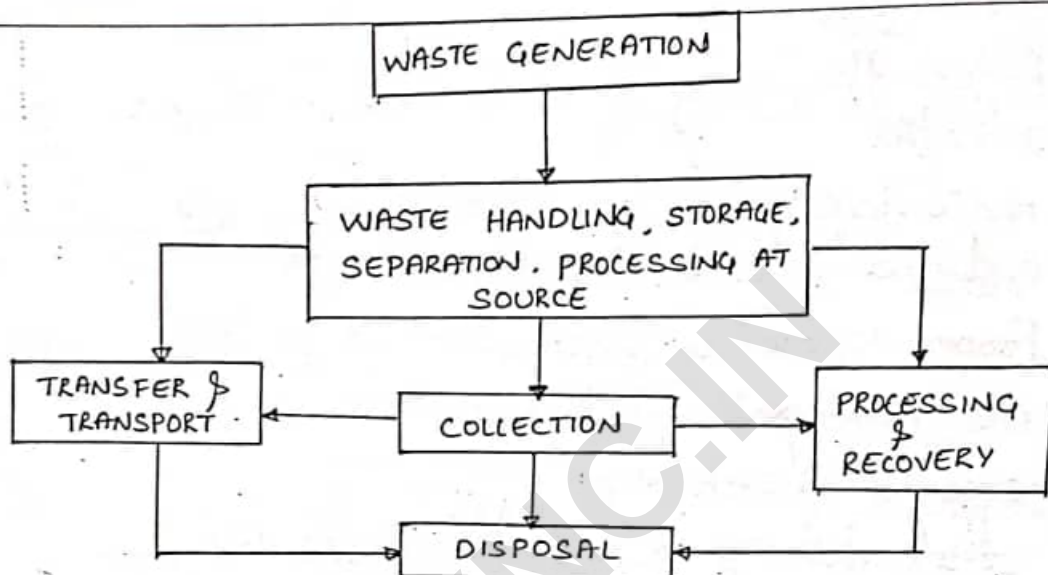
- * Make people aware about concept of reduce, recycle and reuse
- * Reduce the use of pesticides and fertilisers in agricultural activities
- * Avoid buying packages items as they will lead to garbage and end up with land fills site.
- * Proper disposal of garbage
- * Use biodegradable products.
- * Organic agriculturing
- * Create dumping ground ^{away} from residential areas.

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SCOPE OF SWM & IMPORTANCE OF SWM

- * To facilitate adequate facilities to treat, recover and dispose waste in proper manner.
- * It includes all administrative, financial, legal, planning and engineering functions involved in solutions to all problems of solid waste.
- * To minimise rate of waste generation through education and source reduction
- * To decrease volume of residual waste
- * To efficiently manage all waste generated from point of generation through ultimate disposal.
- * To collect generated waste
- * To process waste in systematic way
- * To make proper utilisation of recovered materials
- * To dispose off waste in most economic way
- * To avoid soil and water pollution
- * To protect public health and natural environment

* FUNCTIONAL ELEMENTS OF SOLID WASTE MANAGEMENT



WASTE GENERATION

- * This consists of waste generated from activity of all community
- * This activity is not controllable
- * In future more control will be exercised over a generation of wastes.

WASTE HANDLING, STORAGE, SEPARATION, PROCESSING

- * It includes storage after generation in to containers
- * Separation of components at source of generation is important.
- * It is important to recycle and reuse the waste at the source of generation.

COLLECTION

- * Includes gathering of solid waste and recyclable material, but also transport of material to location after collection (to land fill site)

SEPARATION, PROCESSING & RECOVERY

- * Separation and processing of solid waste components
- * Recovery of separated materials
- * Transfer of reprocessed materials to transfer stations.
- * Separation is done by
 - screening
 - Manual separation.
- * Processing involves reducing the volume and weight of waste to be disposed and generating energy from it.

TRANSFER & TRANSPORT

2 STEPS

- 1) Transport of waste from smaller collection vehicle to larger.
- 2) Subsequent transfer of wastes over a long distances to processing or disposal site. (motor vehicles, rail cars, barges)

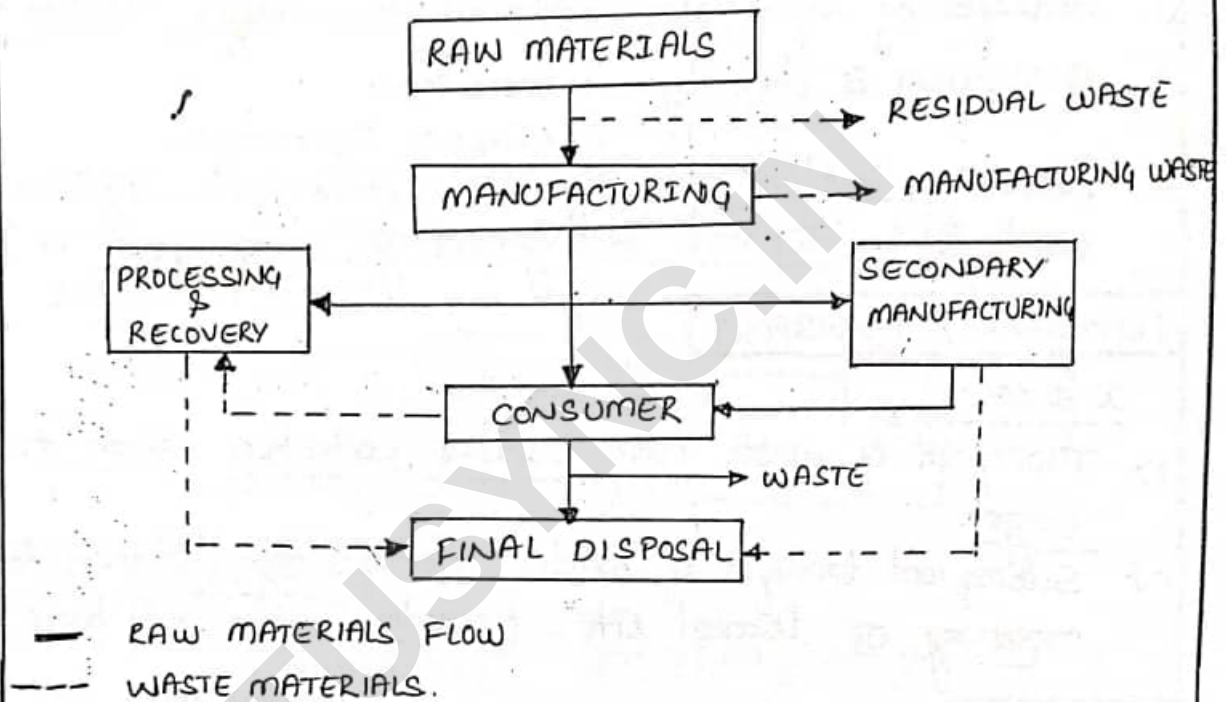
DISPOSAL

- * Scientific disposing of all waste generated by suitable disposal method. Like
 - Land filling
 - Incineration
 - Pyrolysis without causing nuisance to public health and environment.

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MATERIALS FLOW & WASTE GENERATION IN A SOCIETY

Following figure shows how and where solid wastes are generated in the society.



- * Solid wastes are generated at the start of process, starting with minimising of materials
- * Solid wastes are generated at every step in a process as raw materials are converted to goods for consumption
- * Therefore reduction of solid waste generation can be done by reducing consumption of raw materials
- * Solid wastes generation can be reducing by increasing rate of recovery and reuse of materials

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SOURCES OF SOLID WASTES

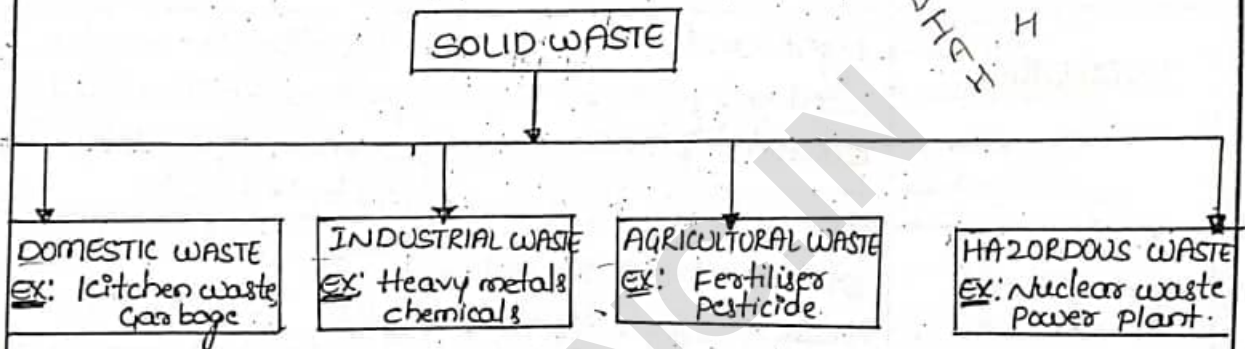
SOURCE	TYPICAL WASTE GENERATORS	TYPES OF SOLID WASTES
RESIDENTIAL	Single and multifamily dwellings	Food wastes, paper, plastic, wood, glass, metals, ashes, and house hold hazardous waste
INDUSTRIAL	Light and heavy manufacturing, fabrication, power, chemical plants	House keeping wastes, food wastes, packaging, construction & demolition materials, etc...
COMMERCIAL	Stores, hotels, restaurants, market, office buildings etc.	Paper, cardboard, plastic, wood, food wastes, glass, metals, hazardous wastes
INSTITUTIONAL	Schools, hospitals, govt. centres	Same as commercial
CONSTRUCTION & DEMOLITION	New construction sites, road repairs, demolition of bldgs.	Wood, steel, concrete, dirt etc...
MUNICIPAL SERVICES	Street cleaning, landscaping, park, beaches, water & waste water treatment plants	Tree trimmings, general wastes from parks, sludge etc...
PROCESS (MANUFACTURING)	Heavy and light manufacturing, refineries, chemical plants, power plants, mining etc...	Scrap materials.
AGRICULTURE	Crops, orchards, vineyards, dairies, farms	Spoiled food wastes - pesticides etc...

CLASSIFICATION OR TYPES OF SOLID WASTES

Solid wastes can be classified on the following three basis

- 1) on the basis of their origin
- 2) on the basis of nature of material present
- 3) on the basis of kind of material present.

1) ON THE BASIS OF ORIGIN



DOMESTIC WASTE :

- * The amount of waste depends on public habits
- * Production of waste in metropolitan cities is higher than towns and villages
- * Average MSW production is about 0.33 kg/capita/day in India
Municipal solid waste

INDUSTRIAL WASTE :

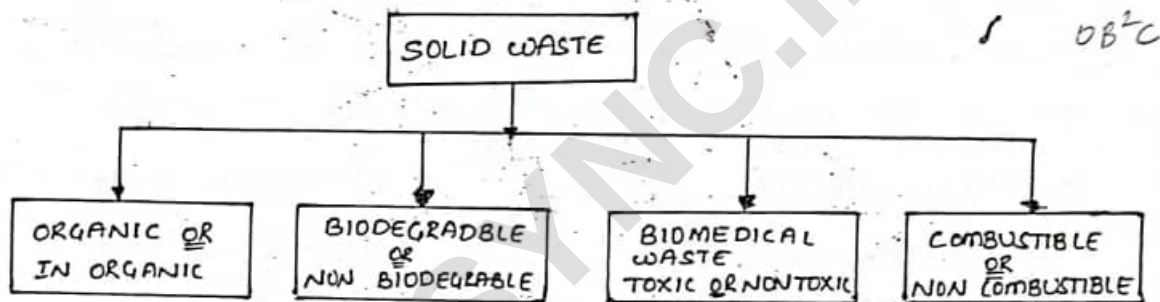
- * Huge amount is produced by different industries
- * Usually it is 10% of total waste generated and is generally in liquid form

AGRICULTURAL WASTE :

- * Main sources of agricultural wastes are wheat, straw, paddy, sugar cane etc...
- * Among animal waste cattle dung, dead animals, urine etc...
- * Agricultural wastes are organic in nature

HAZARDOUS WASTES :

- * Have properties like - corrosivity, ignitability, high reactivity
- * From - Nuclear power plant
Thermal power plant
Hospitals
Labs etc...
- * Not mixed with municipal solid wastes

ON THE BASIS OF NATURE OF MATERIALSORGANIC OR INORGANIC:

Organic compounds are carbon based substances having less percentage of hydrogen and oxygen.

- * EX: organic fertilisers, dyes, plastics etc...
- * Inorganic compounds have very less or no carbon.
EX: lead, iron, mercury, nickel etc...

COMBUSTIBLE/NON COMBUSTIBLE:

- * combustible materials can be burnt easily.
EX: Papers, wooden pieces etc...
- * Non combustible materials that can not be burnt at the ordinary temperatures ($700-1000^{\circ}\text{C}$)
EX: Tin, glass, bottles etc...

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BIODEGRADABLE / NON BIODEGRADABLE :

* Biodegradable are the one which can be easily degraded by microbes in the environment.

Ex: Garbage, paper, cardboard etc..

* The waste which cannot be easily bio-degradable using microbes is called non-biodegradable

Ex: glass, plastic, heavy metals etc..

BIOMEDICAL WASTE:

* All wastes which come from oxidised like residues from operation theaters, sharp needles, springs, blood stained cloths, cotton etc.. is called as biomedical waste.

* It can be further classified as

1) TOXIC WASTES: Is a one which poses great risk to environment because of its contagious & infectious nature. causes diseases if not handled properly.

2) NON TOXIC: It includes non toxic wastes coming from hospitals like syringes, glass bottles, needles etc..

The hospital waste or biomedical waste has to be handled according to local state pollution control board norms

ACCORDING TO KIND OF MATERIALS PRESENT

According to nature of materials they are classified as:

1) FOOD WASTE (GARBAGE) :-

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TYPE OF SOLID WASTE	DESCRIPTION	SOURCES
01 Food waste (Garbage)	waste from cooking, sale of w/meat and vegetables food market	House holds, institutional and commercial centres like hotels, stores, restaurants, markets etc..
02 RUBBISH	<u>combustible</u> paper, card boards, wood boxes, plastic leather, rubber, glass, leaves <u>Non combustible</u> : stone, bricks, ceramics, tin cans etc...	- " -
03 ASHES & RESIDUES	Residues from fires used cooking, thermal power plants, cinders etc..	- " -
04 BULKY WASTES	Auto parts, tyres, refrigerators, furnitures	- " -
05 STREET WASTE	street sweeping, dirt, leaves etc...	vacant plots, streets, side walls.
06 CONSTRUCTION & DEMOLITION WASTE	Roofing, shuttering scrap broken concrete, pipes, wires etc..	construction & demolition sites
07 INDUSTRIAL WASTE & SLUDGES	from industrial process & manufacturing like food processing, boilers house, wood, plastic & metal scraps sludge from STP	Factories, power plants, treatment plants etc..
08 HAZARDOUS WASTE	Explosives, radio active materials, toxic waste	Households, hospitals, stores, industry etc..
09 HORTICULTURAL WASTE	Tree trimmings, leaves wastes from parks & gardens	parks, gardens, roadside trees etc..

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